Bacterial growth curve analysis

A Python project that fits a growth model to bacterial growth curves and compares two incubation temperatures.

The data in this project is simulated. I generated OD600 readings for two conditions so the method could be built and tested end to end. The same code works on real plate-reader data: swap `simulated_od600.csv` for your own file with the columns `condition`, `replicate`, `time_h` and `od600`.

Question

Does a higher incubation temperature (37 C vs 30 C) change how fast a bacterial culture grows, and by how much?

Method

1. **Simulate readings.** Three replicate wells per condition, one OD600 reading every 30 minutes for 24 hours, with realistic well-to-well variation and reading noise.
2. **Fit a model.** Each replicate is fitted with the modified Gompertz model (Zwietering et al., 1990) on ln(OD / OD at time 0), using `scipy.optimize.curve_fit`.
3. **Extract growth parameters.** Lag time, maximum specific growth rate, doubling time (ln 2 divided by the growth rate) and maximum OD.
4. **Compare conditions.** A Welch t-test on the growth rates of the three replicates in each condition.

Results

Condition	Growth rate (per h)	Doubling time (h)	Lag time (h)	Max OD600
37 C	0.603 (SD 0.030)	1.15	1.49	1.57
30 C	0.371 (SD 0.013)	1.87	2.42	1.43

The model fits every replicate closely (R squared about 0.999). The Welch t-test on growth rate gives t = 12.24, p = 0.0018, so the 37 C cultures grow significantly faster than the 30 C

cultures in this simulated experiment.

Limits

- The data is simulated, so the result shows the method works. It is not a biological finding.
- Three replicates per condition is a small sample. With real data I would use more replicates and check the t-test assumptions.
- The Gompertz model is one of several options. Logistic and Baranyi models are common alternatives, and comparing them is a natural next step.

Files

File	What it is
`growth_curve_analysis.py`	The full analysis, from simulation to figure
`simulated_od600.csv`	The simulated readings
`fit_results.csv`	Fitted parameters for every replicate
`growth_curves.png`	Figure: curves with fits, and the growth rate comparison

Run it

```
pip install numpy pandas scipy matplotlib
python growth_curve_analysis.py
```

The random seed is fixed, so you will get the same numbers every time.